

DRAEGER POLYTRON 2 XP

Combustible Gas Detector ? Operation & Maintenance Manual
EXCERPT: Chapter 4 ? Calibration, Maintenance & Troubleshooting
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NOTE: This is an excerpt for training and maintenance purposes.

Full manual available from Draeger Safety AG, Lübeck, Germany.

4.1 INTRODUCTION TO CALIBRATION

The Polytron 2 XP uses a catalytic bead (pellistor) sensor to detect combustible gases. Accurate detection depends on periodic calibration to compensate for sensor aging, poisoning, and environmental factors. Failure to calibrate on schedule is the leading cause of false negatives (failure to alarm when gas is present) and false positives (nuisance alarms).

4.2 CALIBRATION INTERVALS

4.2.1 Standard Calibration Schedule

- Zero check (bump test): Every 30 days, or before any hot work in the monitored area.
- Full calibration (zero + span): Every 6 months, or after any alarm event.
- Sensor element replacement: Every 24 months, or when span calibration cannot be achieved within $\pm 5\%$ of reference gas value after two consecutive attempts.

4.2.2 Accelerated Calibration Triggers

The following conditions require IMMEDIATE calibration regardless of schedule:

- (a) Any alarm event above 25% LEL.
- (b) Sensor exposure to silicone vapours, lead compounds, halogenated solvents, or H₂S concentrations above 200 ppm ? these compounds cause catalytic bead poisoning and permanently reduce sensor sensitivity.
- (c) Sensor readings that drift more than $\pm 3\%$ LEL from known zero in a clean atmosphere.
- (d) Any physical impact or moisture ingress to the sensor housing.

4.3 CALIBRATION PROCEDURE

4.3.1 Pre-Calibration Safety Requirements

IMPORTANT: Before performing calibration in a live area:

- (a) Obtain a Hot Work / Gas Work permit from the area supervisor.
- (b) Verify the area reading is below 10% LEL using a portable gas detector BEFORE connecting calibration gas or opening the sensor housing.
- (c) Inform the shift operator and control room of planned calibration activity.
- (d) Have a second person present as safety observer.

4.3.2 Zero Calibration

- Step 1: Connect certified zero gas (synthetic clean air, <1 ppm HC) at 0.5 L/min.
- Step 2: Allow reading to stabilise (minimum 2 minutes).
- Step 3: If reading is within $\pm 1\%$ LEL of zero, no adjustment required. Log and sign off.
- Step 4: If reading is outside $\pm 1\%$ LEL, adjust zero potentiometer per Figure 4-3.
- Step 5: Record pre-adjustment and post-adjustment readings on Calibration Record Form CF-01.

4.3.3 Span Calibration

- Step 1: Use certified reference gas at 50% LEL methane in nitrogen balance.
IMPORTANT: Reference gas cylinder must have a valid calibration certificate dated within the past 12 months from an accredited laboratory.
- Step 2: Connect at 0.5 L/min via the sensor inlet cap (Figure 4-4).
- Step 3: Allow reading to stabilise (minimum 3 minutes).
- Step 4: Adjust span potentiometer until display reads 50.0% $\pm 1.0\%$ LEL.
- Step 5: If span cannot be achieved within $\pm 5\%$ of reference after two attempts, sensor element requires replacement (see Section 4.5).

4.4 RECOGNISING PRE-FAILURE DRIFT

4.4.1 Early Warning Signs

Sensor degradation typically follows a predictable pattern:

- Phase 1 (Early): Zero drift increases gradually. Span calibration achievable.
- Phase 2 (Mid): Zero drift >2% LEL. Span requires multiple adjustments.